SWINGERsim

Open questions and future work

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Contents

1 Purpose of this document **1**

2 Predictions waiting for observations **2**

3 Open questions from the tests **4**
3.1 Individual systems 4
3.2 Physics and methods 6

4 Possible extensions of the software **7**

5 Research and student projects **8**
5.1 Projects with SQUADRA data 8
5.2 Questions of populations and theory 8
5.3 Student projects 9

6 Questions for colleagues **11**

References **11**

1 Purpose of this document

The User Guide and the Physics and Methods volume describe what SWINGERsim is and how it works. The four test reports describe what it reproduces:

- Part I: published benchmarks and catalogue systems;
- Part II: BU CMi, TIC 120362137 and V994 Her;
- Part III: V907 Sco, QX Cas and AH Cep;
- Part IV: CzeV343, CzeV4315 and the convergence of the fits.

This document collects what those reports leave open. Its sections are:

1. **Predictions waiting for observations** – statements about the future or about data not yet taken, with the data that will confirm or reject them (Section 2).
2. **Open questions** – disagreements and weak points found in the tests, each with what would settle it (Section 3).
3. **Possible extensions of the software** – what SWINGERsim does not do today and what an extension would need (Section 4).
4. **Research and student projects** – questions SWINGERsim is ready for (Section 5).
5. **Questions for colleagues** – items that need data or expertise from outside the project (Section 6).

Each item names the document and chapter where the evidence is described. Nothing here is a claim beyond those reports; this is a working list.

2 Predictions waiting for observations

A prediction is useful only if it says which observation would prove it wrong. **Table 1** lists the predictions of the test reports with that observation. Columns:

- *system*;
- *prediction*: what SWINGERsim says, with its uncertainty where one exists;
- *how to test it*: the observation that would confirm or refute it;
- *where*: the report and section with the evidence.

Dates are calendar years. Ranges are 16–84 % of an ensemble or 95 % of a posterior, as stated.

Table 1. Predictions of the test reports.

system	prediction	how to test it	where
V907 Sco	the present eclipses end in 2052.5–2056.8 (95 %); until then the depths follow the predicted inclination history	depths of the eclipses in future TESS sectors and ground photometry, compared with the model depths	Part III, V907 Sco
V907 Sco	the projected spin–orbit angle λ of the B stars (Rossiter–McLaughlin effect during the eclipses until ~2052) depends on their rotation and history: if tides have damped the spin into its Cassini state, the obliquity is about 7° and |λ| stays below about 7°; a spin aligned in 2021 without dissipation reaches λ ≈ −20° in 2035. Near synchronous rotation the obliquity can change by up to 80–100° in 600 yr	spectroscopy during the eclipses (RM effect), v sin i	Part III, V907 Sco
V907 Sco	light centre moves by about 0.3 mas on the 142-d orbit	Gaia DR4 epoch astrometry (an orbital solution, or an upper limit that would lower $m_3/M - l_3$)	Part III, Gaia
QX Cas	the binary keeps its “heartbeat” brightening near periastron, about 1 ppt at the 2nd and 3rd harmonics of 6.004 d	every new TESS sector; a fit of its shape constrains the present ω and inclination	Part III, QX Cas

system	prediction	how to test it	where
QX Cas	the eclipses ended in 2000–2006.5 and return in 2072–2130 (16–84 % of an MCMC of the dynamical fit with the inclination history, 300 integrated samples; 12 of 300 do not return before 2150). The times of minima do not fix the outer period (χ^2 is flat from 270 to 400 d), so the date is uncertain by decades	not testable soon; DASCH plates test the past (deep eclipses before 1943 or not)	Part III, QX Cas
AH Cep	the eclipses stay constant: the 70-yr third body changes the inclination by only 0.02° in 400 yr	TESS depths over the next sectors; a measured change would require a closer fourth body	Part III, AH Cep
AH Cep	the proper motion differs between Hipparcos and Gaia by about 1 mas/yr	the Hipparcos–Gaia catalogue of accelerations (Brandt 2021)	Part III, Gaia
TIC 120362137	third-body events about every 25 d through 2027, e.g. BJD 2460559, 2460587, 2460610, 2460638 … 2461480	TESS sectors observed since 2024 and new ground photometry at the predicted dates	Part II, TIC 120362137
CzeV4315	binary A keeps eclipsing with a constant inclination (all weighted members change it by < 0.22°/yr)	depths and light-curve inclinations of A in new data	Part IV, CzeV4315
CzeV4315	the third star around B probably has $m_5 > 2$ $M_\odot$ (74 % of the weight), i.e. similar to the stars of B	spectra (a third set of lines); the amplitude of the 119-d timing signal of B (up to about 3 min for 3 $M_\odot$); 119-d radial-velocity variation of B up to about 35 km/s	Part IV, CzeV4315
CzeV4315	the eclipses of B change regime within about 30–60 yr (16–50 %)	long-term monitoring; the inclination change of B should eventually slow down (it is a precession, not a linear drift)	Part IV, CzeV4315

system	prediction	how to test it	where
CzeV343	the fate of binary A depends on the stellar tracks: with MIST, 59 % of 300 futures end in a common-envelope merger and 14 % of the quadruples become dynamically unstable (1.30–1.63 Gyr); with SSE on the same draws, 82 % leave a stripped donor (separation > 50 $R_\odot$ in 83 %) and 75 % become unstable (1.25–2.00 Gyr)	not observable for this system; a population test: stripped giants in 2+2 quadruples with strongly inclined outer orbits	Part IV, CzeV343
V994 Her	a companion of pair C that explains its apsidal motion (U = 214 ± 31 yr) has 0.13–0.31 $M_\odot$ on an 80–182-d orbit (16–84 % of the ensemble members still allowed by the existing eclipse times of C) and shifts the eclipses of C by up to about 2 min	dense timing of C over at least two such periods	Part II, V994 Her

3 Open questions from the tests

Each item states the finding, why it matters, and what would settle it. The first group concerns individual systems, the second the physics and methods.

3.1 Individual systems

QX Cas: the old eclipses and the stellar parameters (Part III, QX Cas)

- *Finding:* With the adopted radii (3.5 and 5.1 $R_\odot$) and temperatures, the photographic minima of 1933–1942 need an inclination of at least 76–78°. A fit that enforces this keeps the eclipses detectable until 1999.0, while none were seen in the mid-1990s. The depths of 1960 and 1980 need 13–47 % of third light, more than a single value can supply.
- *Why it matters:* The dates of the eclipses, past and future, depend on it.
- *What would settle it:*
  - a joint fit of the radii, the light ratio and the third light together with the old depths and the TESS heartbeat light curve;
  - the DASCH plates of 1900–1950;
  - the unpublished light-curve solutions of Bonaro et al. (2009).

QX Cas: a range of dates (Part III, QX Cas)

- *Finding:* A random ensemble weighted afterwards keeps only one or two systems consistent with the whole inclination history. An MCMC of the dynamical fit with the

inclination history as data has now been run (24 walkers × 6000 steps, 30 h). The eclipses ended in 2000–2006.5 and return in 2072–2130 (16–84 %, 300 integrated posterior samples).

- *Why it is still open:* The times of minima do not fix the period of the third star. χ^2 is flat from 270 to 400 d (the best sample in every 20-d bin lies within 3 of the optimum), and the autocorrelation time of the chain is 311 steps, so even 6000 steps are only 9.6 τ after the burn-in. The median return date moved from 2096 (1500 steps) to 2110 (6000 steps). More computing does not fix this; the data do not contain the information.
- *What would settle it:* An independent measurement of the outer orbit: radial velocities of the third star or of the binary over a few hundred days, or Gaia epoch astrometry.

V907 Sco: Gaia sees less motion than modelled (Part III, Gaia)

- *Finding:* The RUWE is 0.81, whereas the model moves the light centre by about 0.3 mas.
- *Why it matters:* It is the only independent check of m_3 and of the light of the third star.
- *What would settle it:* Gaia DR4 epoch astrometry. Until then, a simulation of the Gaia scanning of this source would show whether 0.3 mas must raise the RUWE.

AH Cep: inclination and third light (Part III, AH Cep)

- *Finding:* TESS prefers $i \approx 72.5°$ with about 5 % third light. The published value is 69.8°, and the spectra allow less than 2 % third light.
- *What would settle it:* An independent light-curve analysis, for example with PHOEBE, of the same TESS sectors. Also the Hipparcos–Gaia acceleration for the 70-yr orbit.

BU CMi: the classical apsidal advance (Part II, BU CMi)

- *Finding:* With the apsidal advance that stellar models predict (0.56 and 0.33°/yr), the fit moves the outer eccentricity by 2.6σ. Without it the fit agrees with the paper. A fit with the apsidal constants free and a χ^2 profile in k_2 of pair A (errors held fixed) give $k_{2,A} \leq 0.0012$ ($\Delta\chi^2 = 1$) and ≤ 0.0023 ($\Delta\chi^2 = 4$), against 0.0042 of the models ($\Delta\chi^2 = +13.5$ there, with $k_{2,B}$ pushed to its lower bound). The two pairs trade off ($k_{2,A} + k_{2,B} \approx 0.0045$). Freeing the constants moves e_{out} only from 0.2631 to 0.2663 (published 0.2728 ± 0.0037).
- *Why it matters:* The data want a weaker tidal and rotational apsidal advance than models of synchronously rotating stars give. Lower k_2, slower (sub-synchronous) rotation or missing physics would each do it; this is a tension, not a measurement of k_2.
- *What would settle it:* Projected rotational velocities ($v \sin i$) of the four stars, which fix the rotational part of the advance, and apsidal constants from models of the actual stars.

V994 Her: pair C (Part II, V994 Her)

- *Finding:* The published apsidal rate of pair C needs apsidal constants 1.8 times larger than stellar models give, and the A–B eccentricity is 0.51–0.59 from the minima against 0.69 published.
- *What would settle it:* A closer, unknown companion of C would explain the first point. An ensemble of 3000 companions (0.1–2 $M_\odot$, 8–3000 d) shows which ones give the measured apsidal period. The existing 56 primary minima of C limit a timing signal of 80–400 d to below 2.4 min (2σ, median over the period), and that excludes 55 % of those companions. The rest have 0.13–0.31 $M_\odot$ on 80–182-d orbits. The highest peak in the times of minima (80.9 d, 2.0 ± 0.5 min) is not a detection: aliases at 49–84 d are almost as high. Dense timing of C over at least two such periods is the test. The crowding correction of TESS sector 26 must be checked for the depths.

CzeV343: how often does binary A widen? (Part IV, CzeV343)

- *Finding:* COMPAS (Pejcha et al. 2022) widens binary A in $\gtrsim$ 85 % of realisations, SWINGERsim with MIST tracks in 10–29 %. The same 300 draws run with SSE tracks (the Hurley et al. formulae on which COMPAS is built) leave a stripped donor in 82 % of them and a separation above 50 $R_\odot$ in 83 %; the outcome of the two track families agrees in only 27 % of the draws. With SSE 75 % of the quadruples become unstable, with MIST 14 %.

- *What remains:* The difference lies in the stellar tracks (the response of the donor and so the stability of the mass transfer), not in the dynamics. Which family is closer to the truth needs a detailed binary-evolution code such as MESA on identical inputs.

V907 Sco: the spin of the B stars (Part III, V907 Sco)

- *Finding:* The nodal precession of the binary (63 yr) and the spin precession of its stars (~36 yr at synchronous rotation) are nearly equal: $\eta = g/\alpha \approx 0.58$ against the bifurcation of the Cassini states at 0.587. 600 N-body realisations with free rotation periods and initial obliquities show obliquity changes of 81–82° (median) over 600 yr for rotation periods of 3.5–4.5 d, 13° at 0.5–1 d and 42–44° at 10–30 d. If tides have damped the spin into its Cassini state (6.7°; synchronisation time 9–940 Myr against an age of 400 Myr), the obliquity is small and constant.
- *What would settle it:* The Rossiter–McLaughlin effect during the eclipses, which last until about 2052, and $v \sin i$ of the two stars.

CzeV343: is it in the 3:2 resonance? (Part IV, CzeV343)

- *Finding:* No. A scan of 141 periods of binary A around the exact commensurability (1000 yr each) finds the 3:2 Lindblad resonance of Tremaine (2020), with the angle $3\lambda_A - 2\lambda_B - \varpi_A$ and a width of about 10^{-5} d in P_A. The present system lies 7.9×10^{-4} d, about 80 widths, away. None of its satellites split off by the outer orbit ($k = -6 \ldots + 6$) is active either. Why 2+2 quadruples cluster near such ratios (Kolář et al. 2025) remains open.
- *What would settle it:* A capture experiment with slow migration through the resonance.

CzeV4315: the mass of the third star of B (Part IV, CzeV4315)

- *Finding:* The mass is degenerate with the eccentricity and inclination of its orbit (1.4–4.9 $M_\odot$, 16–84 %).
- *What would settle it:*
  - the amplitude of the 119-day timing signal, or radial velocities of B;
  - a fit of the eclipse times of B with the 2+(2+1) model, which SWINGERsim can do once the minima are available.

3.2 Physics and methods

- **Apsidal constants of stripped and evolved stars.** The future of CzeV343 hinges on k_2 of a stripped giant: about 10^{-3} by our two-zone estimate. SWINGERsim takes k_2 from tables or polytropes for normal stars. For stripped donors, giants with thin envelopes and mass gainers, a structure model, for example from MESA profiles (the `mesa-reduce` tool), is the reliable source.
- **MA01 vs V22 near the boundary (answered for one configuration).** At the onset of instability in CzeV343 the criterion of Mardling & Aarseth gives 1.00, that of Vynatheya et al. 1.19–1.24. 1200 N-body realisations of CzeV343 after the mass transfer, followed for 200 outer orbits and the 243 doubtful ones for 10^4 outer orbits, give the answer (Part IV, CzeV343). Below a Mardling–Aarseth ratio of 1, 81, 59, 33 and 11 % of the members break up in the bins 0.6–0.7 … 0.9–1.0; above 1.1 none does. The Mardling–Aarseth criterion finds 98 % of the break-ups. That of Vynatheya et al. (2022) finds 51 % at its threshold 1 and works best at about 1.17. The neural network of Vynatheya et al. (2023) finds 25 %, and only 32 % even of the systems its own training test (a ghost orbit) calls unstable. These systems lie just outside its training range (mass ratio 10.7 > 10).
- **Convergence of MCMC fits.** The 1000-step chains are about 19 autocorrelation times long; the half kept after the burn-in is about 9.4 τ. For QX Cas, whose outer period the data do not fix, τ is 311 steps, so even 6000 steps are only 9.6 τ after the burn-in (see above). An MCMC also checks the least-squares fit: for V907 Sco its best sample has a χ^2 2.3 below the optimum (with the final jitters of the fit). A least-squares fit started at that sample, with the jitters re-estimated as in Part III, returned to the same optimum (χ^2 222.73, every parameter within 0.03σ), so the optimum does not depend on the start. $\Delta\chi^2$ = 2.3 lies far inside the 1σ region of 14 parameters ($\Delta\chi^2 \approx 16$): the valley is flat, and fit and MCMC agree. Comparing the lowest sample with the optimum requires removing the term -2 ln e that the ecosw/esinw parametrisation adds to -2 ln p. The

medians are stable, and the widths of the 95 % ranges are less certain than those of the 1σ intervals. About 1100 steps would reach 10 τ after the burn-in (Foreman-Mackey et al. 2013), and about 5300 steps the 50 τ of the emcee documentation.

- **Main-sequence tides.** The equilibrium tide with inertial waves reproduces the circularisation of young binaries. It gives 7.5 d instead of the 12.1 ± 1.2 d of M67 at 4 Gyr (Part I). Stronger dissipation (resonance locking, non-linear dynamical tides) is missing from all published equilibrium-tide models and lies outside the scope of SWINGERsim.
- **Stellar tracks.** SSE and MIST differ by about 10 % in main-sequence lifetimes and radii at its end. When the timing of an event matters, both families should be compared (Part I).

4 Possible extensions of the software

SWINGERsim is complete within its stated scope: hierarchical systems of three to ten stars, from the pre-main sequence to the end of the AGB, with tides, mass transfer, common envelopes, discs and the observables of eclipsing systems (*User Guide*, chapter Introduction). The tests suggest the following extensions. None of them is needed for the results of the test reports.

Table 2. Extensions suggested by the tests. Columns:

- *extension*: what is missing;
- *motivated by*: the test in which it would have helped;
- *effort*: a rough estimate of the work — small (days), medium (weeks) or large (months).

extension	motivated by	effort
fit of stellar radii, light ratio and third light together with historical eclipse depths (not only times)	QX Cas	medium
history-conditioned search: MCMC directly on an inclination history	QX Cas dates	small (existing fit chain, configuration and computing time)
synthetic Gaia astrometry: light-centre track sampled with the Gaia scanning law, RUWE and an epoch-astrometry likelihood for DR4	V907 Sco, QX Cas	medium
Hipparcos–Gaia acceleration as an observable of an ensemble or fit	AH Cep	small
apsidal constants from detailed structure of stripped and evolved stars along a mass-transfer history	CzeV343	medium
a stage-2 integration that continues mass transfer and a common envelope inside the N-body run	CzeV343	large
resonance-angle diagnostics for near-commensurate 2+2 systems	CzeV343 (3:2)	small

extension	motivated by	effort
selection model of surveys (TESS, ZTF) for population statistics of eclipsing multiples	projects E and O below	large

5 Research and student projects

SWINGERsim is meant for modelling specific systems from their data. The projects below use what it already does. Each has a question, the tools it would use and the data it needs.

5.1 Projects with SQUADRA data

Table 3. Projects on systems observed by the SQUADRA group. Columns: the project and its question, what SWINGERsim provides, and what is still needed (usually data).

project	question	SWINGERsim provides	still needed
A. Systems with changing inclination	which outer orbits explain the measured inclination changes (CzeV4315, ASASSN-V J075203 and similar), and when do their eclipses appear or disappear?	ensembles with observation weights and evidences for alternative hierarchies; the dynamical fit of minima, radial velocities and light curves	times of minima of both pairs; light-curve inclinations at several epochs
B. Eclipse forecasts for the SQUADRA list	for each target, the probability that the eclipses of either pair disappear within 10–50 yr: which systems to observe first	configurations from published solutions, ensembles, the worker for batches	a complete table of periods, masses and outer orbits
C. CzeV343 and the 3:2 resonance	is the system in the resonance or passing through it, and how will its O−C change?	apsidal motion, chaos maps (MEGNO), long integrations, the fit chain	resonance-angle diagnostics (Table 2)
D. Compact 2+2 systems with outer periods below a year	secular theory fails; what does a full N-body model of their O−C give?	the dynamical fit (stage 2) with all pairs	minima of the group’s targets

5.2 Questions of populations and theory

Table 4. Wider questions. Columns: project, question, what SWINGERsim provides.

project	question	SWINGERsim provides
E. Alignment of 2+2 orbits	are 2+2 orbits mostly aligned (formation in a disc) or isotropic (dynamical capture), judged from the statistics of doubly eclipsing systems?	eclipse probabilities of populations (validated against Vokrouhlický 2016); missing: a survey selection model
F. The excess of the 3:2 period ratio	why do 2+2 systems show an excess near 3:2 (Kolář et al. 2025)? Capture into resonance during orbital migration, as known for planets?	chaos maps, a parametric gas phase, tides
G. Mergers and contact binaries from ZLK	how many 2+2 systems end as mergers, contact binaries or mass-transferring binaries, and when?	ZLK with tides, stellar evolution, mass transfer, common envelope, contact binaries; Part IV is one example
H. Instability after mass transfer	how often does mass transfer destabilise a quadruple, and what follows: merger by ZLK, as in CzeV343, or ejection?	the two-stage method of Part IV
I. Formation by binary–binary scattering	cross-sections for forming stable 2+2 and 3+1 systems in young clusters	scattering experiments, escape theory, tidal capture
J. Gas phase and alignment	which distribution of mutual inclinations does a gas phase leave behind?	parametric gas phase
K. Pulsating stars as clocks	do the light-time delays of δ Scuti, β Cephei or γ Dor pulsations together with the eclipse times give the masses of the outer orbit without spectroscopy?	pulsations evaluated at the emission time; minima from synthetic light curves
L. Mass transfer in triples	does a tertiary change the outcome of mass transfer (Algols, stripped stars), and does the transfer change the outer orbit and the eclipses?	mass transfer in inner binaries and from a tertiary, with its effect on the other orbits
M. Common envelope in multiples	how often does a common envelope unbind or destabilise the outer orbit?	energy prescription with the response of the outer orbits
N. Young massive multiples	which star of a young O-star quadruple interacts first (QZ Car-like systems)?	MIST to high masses, winds, effective ages (Part I, QZ Car)
O. Hidden multiples	how many eclipsing binaries in TESS, ZTF and ASAS-SN start or stop eclipsing within the survey baseline?	eclipse appearance statistics of a population

5.3 Student projects

Table 5. Student projects. Columns: level, topic (with the project letter of Tables 3–4), task, difficulty.

level	topic	task	difficulty
Bachelor	eclipse forecasts for ten SQUADRA systems (B)	configurations from published solutions, ensembles, a summary of when eclipses vanish	low
Bachelor	binary–binary scattering (I)	scan velocities and orbit ratios, compare with Antognini & Thompson (2016)	low to medium
Bachelor	time delays of a δ Scuti star in a TESS eclipsing binary (K)	measure the delays in TESS, compare with a SWINGERsim model	low to medium
Master	dynamical model of CzeV4315 (A)	fit of the minima of both pairs with the 2+(2+1) model; a paper with the observers	medium
Master	QX Cas: radii, third light and the old eclipses (open question of Section 3)	joint fit of depths, TESS heartbeat and times of minima; DASCH plates	medium
Master	chaos maps and the 3:2 resonance in CzeV343 (C, F)	MEGNO maps, comparison with the resonant secular theory of Breiter & Vokrouhlický (2018)	medium to high
Master	Algols with tertiaries (L)	ensembles of mass-transferring binaries with tertiaries, comparison with known systems	medium
PhD	alignment of 2+2 orbits from the eclipsing population (E)	survey selection model, Bayesian inversion	high
PhD	origin of the 3:2 excess (F)	N-body with migration and tides, comparison with the observed systems	high
PhD	fates of 2+2 systems (G, H)	ZLK with stellar evolution and mass transfer for a population, comparison with population-synthesis codes	high

Ensembles of 10^4 members run in hours on a 12-core computer, so a student can concentrate on the physics.

6 Questions for colleagues

Table 6. Items that need data or expertise from outside the project. Columns: whom to ask, the question, and which open item of this document it serves.

who	question	serves
P. Zasche	CzeV4315: the amplitude of the 119-day timing signal of B; minima files of both pairs for a dynamical fit; is the inclination of A constant in the data? Are spectra possible?	Section 3, CzeV4315; projects A, D
J. Kolář and the SQUADRA group	completion of the SQUADRA target table (periods, masses, outer orbits); review of Parts II–IV	projects B, D
O. Pejcha	CzeV343: a COMPAS run with the inputs of Part IV (or the settings of the published runs), to compare how often binary A widens; interpretation of the ZLK merger channel	Section 3, CzeV343
D. Vokrouhlický	comparison of the N-body results with his secular theory for misaligned 2+2 systems; population eclipse statistics	project E
M. Brož	light curves of multiple systems with PHOEBE (Xitau) as an independent check of AH Cep and QX Cas	Section 3, AH Cep and QX Cas
E. Guinan / the Villanova group	the light-curve solutions of QX Cas of Bonaro et al. (2009) (inclinations, third light, radii)	Section 3, QX Cas

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